Lecture 31.10: Hierarchical Clustering — Bottom-Up Agglomerative Aur Linkage Methods [Hinglish]











📌 Topic Overview

Comprehensive conceptual analysis of Hierarchical Clustering. Contrasts bottom-up Agglomerative clustering against top-down Divisive clustering, and provides mathematical derivations for Single, Complete, Average, and Ward linkage criteria.

📋 Is Topic Mein Hum Kya Seekhenge?

- **Hierarchical Clustering** kya hoti hai: Data ko tree-like structure mein jodna jahan pehle se K batana zaroori nahi hota.
- **Agglomerative (Bottom-Up)**: Har point shuruat mein apna cluster hota hai, fir sabse paas wale do clusters aapas mein judte chale jaate hain.
- **Divisive (Top-Down)**: Saare points pehle ek giant cluster mein hote hain, fir unhe divide kiya jata hai.
- **Linkage Methods** (Do clusters ke beech doori nikalne ke 4 tareeqe): **Single**, **Complete**, **Average**, aur **Ward Linkage**.
- Ward Linkage kyu best aur default method hota hai.

🔍 Har Ek Point Ka Simple & Detailed Explanation

- **Hierarchical Clustering Ka Core Idea**: K-Means mein hume shuruat mein hi batana padta tha ki $K = 3$ ya $K = 4$. Lekin **Hierarchical Clustering** mein hume pehle se K decide nahi karna padta! Yeh poore data ka ek family tree (hierarchy) bana deti hai.

- **Agglomerative (Bottom-Up) Approach (Sabse Popular)**:

- 1. Shuruat mein har ek data point apna khud ka individual cluster hota hai (N points = N clusters).
- 2. Sabse paas wale do clusters ko aapas mein merge kar diya jata hai ($N-1$ clusters bache).
- 3. Yeh merging tab tak chalti rehti hai jab tak end mein ek single giant cluster na ban jaye!

- **Divisive (Top-Down) Approach**: Iska ulta hota hai. Saare points 1 cluster se start hote hain aur recursive splits karke single points tak pahunchte hain.

- **Do Clusters Ke Beech Distance Kaise Nikalein? (Linkage Criteria)**:

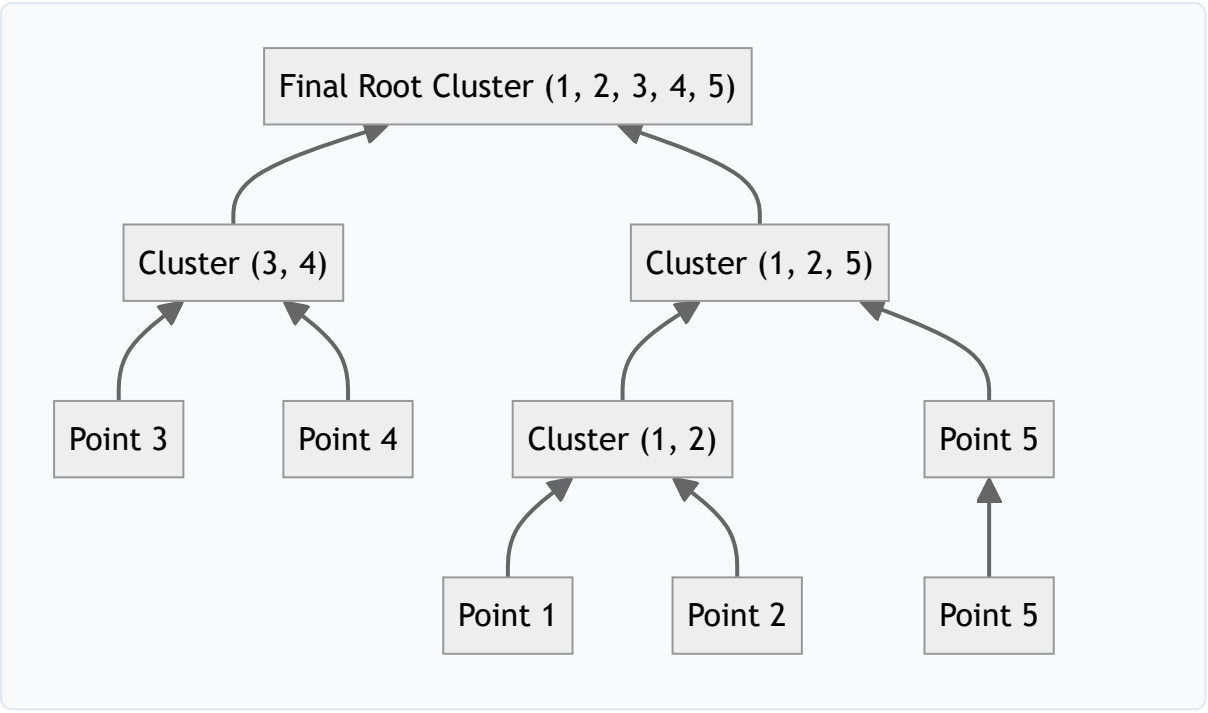
- ◦ **Single Linkage**: Cluster A aur Cluster B ke do sabse paas wale points ka distance: $d_{min}(A,B) = \min d(x,y)$. (Chaining effect ka khatra).
- ◦ **Complete Linkage**: Do sabse door wale points ka distance: $d_{max}(A,B) = \max d(x,y)$. (Compact, spherical clusters banata hai).
- ◦ **Average Linkage**: Dono clusters ke saare pairs ka average distance.
- ◦ **Ward Linkage (Default in Scikit-Learn)**: Do clusters ko merge karne par variance (sum of squared errors) mein kitna izafa hoga: ΔVar. Jo merge variance ko sabse kam badhaye, usi ko merge kiya jata hai!

- **Pure Theory Lecture**: Is lecture mein ma'am ne blackboard par Agglomerative vs Divisive aur Linkage formulas explain kiye hain, koi code nahi hai.

📋 Linkage Criteria Comparison Matrix

Linkage Method	Mathematical Definition	Cluster Shape Tendency	Outlier Sensitivity
Single Linkage	$d(A,B) = \min_{x \in A, y \in B} d(x,y)$	Can follow elongated shapes	High (prone to chaining noise)
Complete Linkage	$d(A,B) = \max_{x \in A, y \in B} d(x,y)$	Compact, equal diameter spheres	Low (sensitive to outer boundaries)
Average Linkage	$d(A,B) = \frac{1}{\|A\|\|B\|} \sum d(x,y)$	Balanced variance clusters	Moderate
Ward's Linkage	$\Delta \text{ESS} = \frac{\|A\|\|B\|}{\|A\|+\|B\|} \|\mu_A - \mu_B\|^2$	Compact, spherical, variance-optimal	**Lowest (Industry standard default)**

📊 Agglomerative (Bottom-Up) Merging Flow



🔢 Mathematical Derivation

Ward's minimum variance criterion for merging clusters A and B:

$$\Delta I_{AB} = \frac{N_A N_B}{N_A + N_B} \|\mathbf{m}_A - \mathbf{m}_B\|^2$$

where $\mathbf{m}_A$ and $\mathbf{m}_B$ are the centroid mean vectors of clusters A and B.

📌 Note on Code

Yeh pure conceptual, mathematical, aur blackboard theory lecture hai. Ma'am ne is lecture mein code nahi likha hai, isliye rule ke mutabiq isme koi extra synthetic code add nahi kiya gaya hai.